

E-Commerce Backend Systems

Abstract

The ShopEasy E-Commerce Website is a standalone web-based shopping application developed using HTML, CSS, and JavaScript. The main objective of the project is to provide users with a simple and user-friendly platform for browsing products, searching and filtering items, adding products to a cart, placing orders, selecting payment methods, and tracking orders. The website contains multiple product categories such as electronics, fashion, home appliances, beauty products, and books.

The system provides features such as user login and registration, product sorting from A–Z and Z–A, discount deals, shopping cart management, checkout, delivery address management, order confirmation, tracking information, order cancellation, and delivery-date modification. The project uses browser local storage to maintain cart, user, and order information without requiring a backend server. The system demonstrates how modern e-commerce functionality can be implemented in a simple standalone web application.

Introduction

E-commerce has become an important part of modern shopping because it allows customers to purchase products conveniently through the internet. Traditional shopping requires customers to physically visit stores, search for products, compare prices, and complete purchases manually. An online shopping system reduces these difficulties by providing products and purchasing facilities through a web interface.

The ShopEasy E-Commerce Website is designed to demonstrate the basic functionality of an online shopping platform. Users can browse different products, search for required items, view discounted products, add items to their shopping cart, provide delivery details, select a payment method, and place an order.

The project is developed as a standalone web application using HTML for structure, CSS for design, and JavaScript for functionality. Local storage is used to save important information such as cart items and orders. This makes the project suitable for demonstrating the basic concepts and workflow of an e-commerce application without requiring a separate database or server.

Scope of the Project

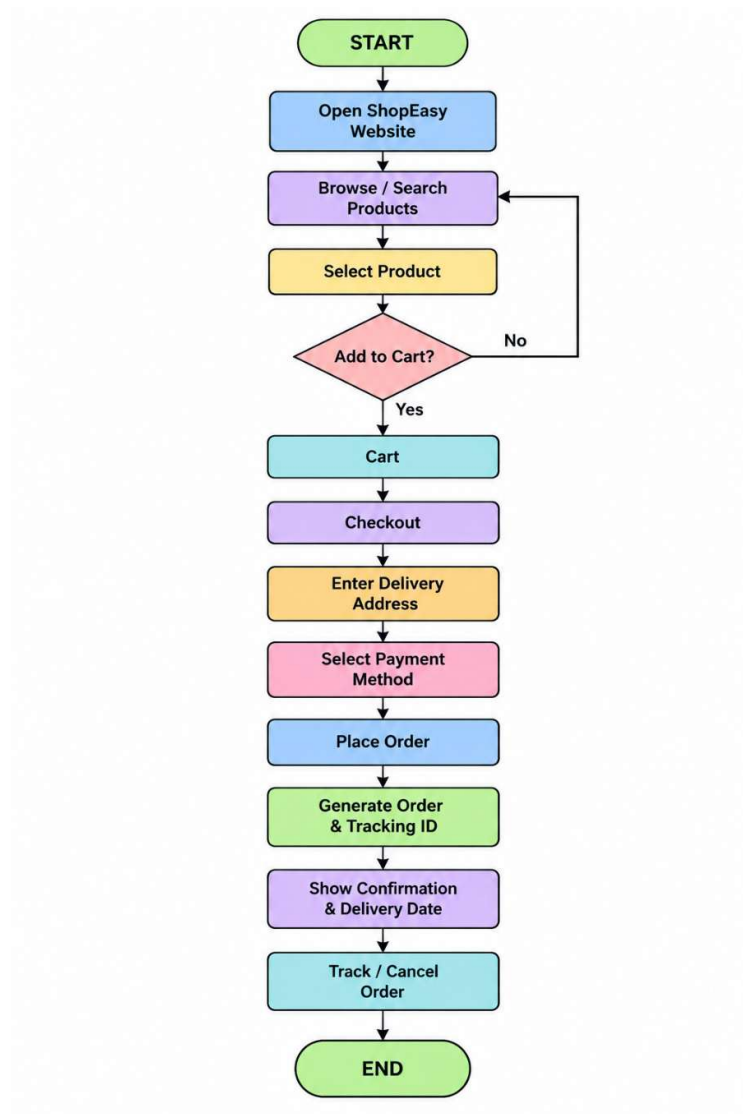
The scope of the ShopEasy E-Commerce Website includes the following features:

- User login and registration.
- Display of multiple products and categories.
- Product search functionality.
- Product filtering by category.
- A–Z and Z–A product sorting.
- Display of discounted products and deals.
- Shopping cart management.
- Increasing or decreasing product quantities.
- Checkout and delivery-address collection.
- Multiple payment-method selections.
- Order placement and confirmation.
- Automatic three-day estimated delivery date.
- Generation of order ID and tracking ID.
- Order history management.
- Order tracking.
- Order cancellation.
- Delivery-date modification.
- Email confirmation through the user's mail application.
- Message/share confirmation.
- Dark and light mode.
- Responsive design for different screen sizes.
- Local storage for maintaining user, cart, and order information.
- Basic rule-based customer assistance.

Limitations

Since the project is a standalone HTML application, it does not provide real online payment processing, a server-side database, or actual server-based email delivery. These features can be integrated in a future version using a backend and appropriate third-party services.

Flow Chart



Conclusion

The ShopEasy E-Commerce Website successfully demonstrates the basic working of an online shopping system using HTML, CSS, and JavaScript. The application provides important e-commerce functions such as product browsing, searching, sorting, deals, cart management, checkout, payment selection, order placement, tracking, cancellation, and delivery management.

The use of local storage allows the application to maintain cart and order information without requiring a backend server. The project provides a simple and user-friendly interface and demonstrates how different components of an e-commerce system work together.

In the future, the project can be enhanced by adding a real backend database, secure user authentication, real payment gateway integration, server-side email notifications, administrator functionality, product reviews, and real-time order tracking.